



PRODUCT NAME

Premium Organic Shower Gel Formula

Description

A cost-effective, biodegradable shower gel formulated to provide gentle cleansing by utilizing organic and natural ingredients without synthetic fragrances. Designed to maintain a balanced pH level for optimal skin health.

Technical Specifications

pH Level:	7
Viscosity:	Medium
Shelf Life:	12 months
Batch Size:	20-100 kg
Processing Time:	2-3 hours
Temperature:	60°C during phase mixing, 30°C for final adjustments
Storage Conditions:	Store in a cool, dry place away from direct sunlight.
Equipment:	Mixing tank with heating and cooling capabilities, pH meter, Homogenizer
Certification:	Cosmos Organic, EcoCert

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Water	Aqua	68.0%	Solvent
2	Decyl Glucoside	Decyl Glucoside	10.0%	Mild Surfactant
3	Coco-Glucoside	Coco-Glucoside	8.0%	Secondary Surfactant
4	Glycerin	Glycerin	5.0%	Humectant
5	Xanthan Gum	Xanthan Gum	1.0%	Thickening Agent
6	Aloe Vera Juice	Aloe Barbadensis Leaf Juice	5.0%	Moisturizing Agent
7	Citric Acid	Citric Acid	0.5%	pH Adjuster
8	Sodium Benzoate	Sodium Benzoate	0.5%	Preservative
9	Natural Colorant/Pea Green Extract	Spinach Extract	2.0%	Natural Colorant
Total			100%	

Manufacturing Process

Phase 1: Aqueous Phase

1. Heat water to 60°C.
2. Disperse Xanthan Gum into heated water while stirring until fully hydrated.
3. Add Glycerin and Aloe Vera Juice to the blend and mix until uniform.

Phase 2: Surfactant Phase

1. In a separate container, blend Decyl Glucoside and Coco-Glucoside.
2. Slowly incorporate the surfactant mixture into the aqueous phase with constant stirring.

Phase 3: Adjustments Phase

1. Cool the mixture to 30°C.
2. Adjust the pH to 7 using Citric Acid.
3. Add Sodium Benzoate as a preservative and ensure dispersion.

Phase 4: Finishing Phase

1. Add Natural Colorant/Pea Green Extract to the mixture.
2. Stir until the mixture is homogenous and smooth.

Required Equipment

Mixing tank with heating and cooling capabilities, pH meter, Homogenizer

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Cosmos Organic, EcoCert

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 20-100 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply a liberal amount onto wet skin during bath or shower. Gently massage into the skin to create a lather, and rinse off thoroughly with warm water.